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BEHAVIOUR AND LEARNING

The story of twentieth-century academic Psychology is the story of an ultimately unsuccessful struggle against an ever more obvious fragmentation. Intelligence and its testing provided an early example of the discipline's tendency to annex new areas without being able to assimilate them. Psychologists had gained an academic foothold by doing experiments on such topics as sensation, perception and memory. For some time, that remained the respectable core of the discipline, but how test intelligence related to this core was far from clear. It was much easier to annex such a field institutionally than to assimilate it intellectually.

That situation was to be repeated many times over in the course of the twentieth century. Child study, or paedology, as it was known in some countries, was another early example. Originating in joint efforts by physicians and educationists, it became transformed into child psychology, rapidly in the US, more slowly in Europe. But its links to core areas of the discipline remained tenuous at best. The same could be said of educational psychology, another early branch. In the period between the two world wars the discipline sprouted as many arms as Shiva, the Hindu deity. A psychological social psychology challenged its sociological rival, 'personality' and 'motivation' emerged as semi-autonomous fields of research and teaching, industrial psychology flourished, clinical psychology became a reality.

What link was there between these fields, except that they all claimed to be 'psychological'? But did that mean anything beyond a vague sense of a common focus that was based on popular images rather than on solid scientific grounds? Grouping these diverse areas together as branches of one discipline undoubtedly had certain practical advantages. It advanced the cause of professionalization by implying that the more practically oriented branches had a respectable link to basic science, and it legitimized the otherwise esoteric interests of the academics by implying that their work had significant practical applications. But, for the most part, such implications were nothing more than promissory notes to be cashed in at some time in the future. Why should anyone accept these notes, and, more importantly, how could psychologists justify such promises to themselves?

In the period under consideration here, that justification depended to a very large extent on the notion that, ultimately, Psychology was *one* discipline whose various branches would turn out to be linked by *one* set of principles. The grouping together of diverse fields of research and practice under one umbrella would then be more than a matter of historical accident and administrative convenience, it would be the logical consequence of deep theoretical links; common scientific 'laws' would unify the discipline. As a first step in this direction, the various parts of the discipline would need to be tied together by common categories of discourse. Such common categories would establish the claim that there were indeed phenomena of importance that were common to all fields of Psychology. Then one could study these common phenomena in order to discover the principles that unified the discipline.

This was the role played by the categories of 'behaviour' and 'learning' in the history of twentieth-century American Psychology.[1] Of the two, 'behaviour' was more foundational, for it became the category that the discipline used to define its subject matter. Whether one was trying to explain a child's answers on a problem-solving task, an adult's neurotic symptomatology, or a white rat's reaction to finding itself in a laboratory maze, one was ultimately trying to explain the same thing, namely, the behaviour of an organism. Classifying such diverse phenomena together as instances of 'behaviour' was the first necessary step in establishing the claim that Psychology was one science with one set of explanatory principles. 'Learning' was not quite as basic a category, but, historically, its role was so deeply entwined with that of 'behaviour' that one can hardly do justice to the one without the other. For several crucial decades they formed an almost inseparable pair, for the 'laws of learning' provided the core example of those behavioural principles that were supposed to unify the discipline. In the present chapter we will trace the origins of this complex, focusing first on 'behaviour' and then on 'learning'.

The history of 'behaviour' is not only intertwined with the history of 'learning', it is also deeply entangled with the history of behaviour*ism*. Unlike the other categories considered here, it has the unique distinction of having given its name to a *movement*. That leads to certain difficulties. We have to be careful not to confuse the history of the movement with the history of the category. They are far from being the same. Historically, behaviourists had no monopoly on the category of behaviour. It was there as a scientific category before they picked it up and nailed it to their masthead, but even then they were not the only ones to give it prominence. One did not have to be a card-carrying member of one of the behaviourist schools to agree to the definition of Psychology as the science of behaviour. Our primary focus here is on 'behaviour' as a category of disciplinary discourse, not on behaviourism as a movement.

That focus is at variance with most of the extensive historiography on the topic of behaviourism (see, however, Kitchener, 1977). By and large, this historiography concentrates on particular individuals self-identified as behaviourists and particular doctrines to which these individuals were explicitly committed. Such a procedure yields valuable historical insights. But it is more helpful in understanding a minority than in following the commitments of the majority of psychologists who, though influenced by behaviourism, did not identify themselves as behaviourists and rejected many of the movement's specific doctrines. If we wish to come closer to an understanding of the aspirations and self-understanding of a discipline, rather than of a movement within it, we will have to focus on its common practices (Danziger, 1990a) and on the categories of shared disciplinary discourse. 'Behaviour' and 'learning' are particularly important examples of such categories. Others will be discussed in Chapter 9.

Five layers of 'behaviour'

In following the vicissitudes of 'behaviour' it is possible to distinguish five periods. Chronologically, the distinction between these periods is not sharp. As one would expect, they merge into each other. Moreover, each period left a kind of sediment behind, so that later on, one encounters layers of usage that are reminiscent of an archaeological site. Each layer has distinct characteristics and its origins can be approximately dated.

Leaving aside older lay uses of 'behaviour', one finds a first layer of relevant scientific use[2] in the context of comparative psychology. At the beginning of the twentieth century, one can detect a new tendency among comparative psychologists to use the phrase 'animal behaviour' when referring to their subject matter. In the year 1900, Lloyd Morgan, then the best-known representative of the field in the English-speaking world, entitled his latest text *Animal Behaviour*. This was a departure from the titles of his earlier books, covering similar topic areas, that had made use of such formulations as 'comparative psychology', 'animal life and intelligence', etc. A few years later, H.S. Jennings (1906), a rising star of American Zoology, entitled his magisterial survey of the field *Behavior of the Lower Organisms*. A paper by Jennings published in 1899 still bore the title, 'The psychology of a protozoan', but by 1904 he had switched to 'The behavior of Paramecium'. During the first decade of the century the term 'animal behaviour' became well established among those working in the area of what had come to be known as comparative psychology. In 1911, a new monograph series in the area was called simply, 'Behavior Monographs', and a new journal began publication under the title *Journal of Animal Behavior*. While the field was obviously expanding, it should be noted that for many practitioners the term 'animal behaviour' was at first meant to complement, rather than displace, such terms as 'comparative psychology' and 'animal mind'. As late as 1930, Lloyd Morgan published a book on *The Animal Mind*, while Margaret Washburn's *The Animal Mind: A Text-Book of Comparative Psychology* went into its fourth edition in 1936. But by then such terminology had become unusual.

A second phase in the use of 'behaviour' begins slightly later but overlaps with the first. At this stage the term makes its entry into general psychology but without the behaviouristic colouring it was to acquire later. The first text to announce in its title that Psychology was 'the study of behaviour' appeared in 1912. Its author was William McDougall, soon to take on the role of one of behaviourism's bitterest foes. In the previous year, 1911, at least two books had been published whose titles referred to 'human behavior' as their subject matter (Meyer, 1911; Parmelee, 1911). While one of their authors could be regarded as a behaviourist before J.B. Watson had proclaimed the label, the other, Parmelee, could not. In the same year, a relatively conventional psychological text (Pillsbury, 1911) introduced its subject matter by defining it as the science of behaviour. By the end of 1912, J.R. Angell, spokesman for the functionalist school of psychology, had delivered a timely address to the American Psychological Association on 'Behavior as a category in psychology' (Angell, 1913).

Just two months later, J.B. Watson, a young comparative psychologist, presented a paper at Columbia University which ushered in a third phase in the modern history of 'behaviour' (Samelson, 1981). Watson (1913) called his paper 'Psychology as the behaviorist views it'. The change was quite drastic. From being a label for a category used in defining the subject matter of Psychology, 'behaviour' was now given the distinction of labelling a certain kind of psychologist. It was elevated above other psychological categories in that it not only referred to an object of investigation but also supplied an *identity label* that challenged psychologists either to accept or reject it. From the point of view of the discipline's history, the most significant aspect of Watson's challenge was that American psychologists took it so seriously. They could have ignored it, or shaken their heads in amused disbelief, as their European colleagues tended to do. But they didn't. For about fifteen years after the end of World War I, the question of the behaviourist identity sharply engaged the attention of the American psychological community. As a result, the category of behaviour acquired a new set of connotations. There was now a considerable gap between what the term meant in psychological and in lay discourse.

It is well known that during the 1930s behaviourism acquired a new lease of life in the form of what came to be known as neo-behaviourism. For the category of behaviour, this entailed a partial change of connotation. During the previous three decades 'behaviour' had acquired a certain substantive content that included a commitment to mechanistic explanation and epiphenomenalism. In the period of neo-behaviourism such commitments were still common among the friends of behaviour, but they had become optional. Instead of being defined by implicit claims about the nature of reality, 'behaviour' is now increasingly defined by the *practice* of psychologists. It refers to whatever objects can be constructed by the standard technology of the discipline. Commitment to the norms governing this technology takes the place of direct and substantive metaphysical commitments. Of course, the latter are still present, but they have become implicit and invisible.

This development prepared the way for the fifth and final stage in the twentieth-century history of 'behaviour'. In the years following World War II the term became associated with the new category of 'behavioural science'. The discipline of psychology had never had a monopoly on 'behaviour' and its investigation, and in the post-war period the use of this category became more widespread than it had ever been before. Although psychological usage remained in some sense exemplary for other disciplines, there was now some reciprocal effect on the discipline of Psychology. In the main, this took the form of reinforcing the direction of change that had been initiated in the previous period. More and more, 'behaviour' became a methodologically grounded category.

This overview of the layers of meaning acquired by 'behaviour' provides a rough framework for a consideration of more specific historical and theoretical issues arising at each stage. Because my main focus here is on the *origin* of categories, I will concentrate on the earlier stages.

Mind as inference

If we trace the roots of the peculiarly twentieth-century use of 'behaviour' we arrive at the animal psychology of the first years of the century. What was involved in this early discourse about 'animal behaviour'? For Jennings, a respected pioneer in this area, 'behaviour' owed its role to its juxtaposition with its antipole, 'consciousness'. He begins his early text *Behavior of the Lower Organisms* by noting that 'assertions regarding consciousness in animals, whether affirmative or negative, are not susceptible of verification' (Jennings, 1906: v). For purposes of scientific study we have to turn to something else, namely, behaviour. This is defined as 'the general bodily movements of organisms', and when we study it, 'we are dealing with actual objective processes' and aspire to 'a knowledge of the laws controlling them' (ibid.). Behaviour, being 'objective' in some essential sense, is different from 'consciousness' which is not.

In drawing a sharp line between consciousness and objectivity Jennings was merely following in the footsteps of a number of European biologists who had turned their attention to studying the reactions of organisms to their environment. In this endeavour they encountered a special problem of language. How was one to describe the reactions of worms or molluscs, for example? Traditional usage offered two alternatives: one could use the language commonly employed in referring to the actions of humans. But that language implied self-conscious agents who pursued moral and rational intentions. As long as the conduct of all living things was seen as reflecting the intentions and wisdom of their divine creator, the use of such language in connection with even the humblest of animals did not seem particularly inappropriate. But once that way of viewing life had been displaced by nineteenth-century natural science, the application of a language of intentionality to the actions of the lower organisms seemed ever more incongruous. The alternative was to employ the physicalistic language commonly used to describe the changes of state and position of inanimate bodies. But to most biologists that did not seem appropriate either. Therefore, by the end of the nineteenth century there was considerable discussion about developing a terminology for studies of animal behaviour that would be 'objective' and yet preserve the special qualities that distinguished animate from inanimate action (Beer et al., 1899; Claparède, 1903; Dzendolet, 1967).

What gave this topic psychological interest and relevance was the fact that most biologists still conceived of the special qualities of animal action in terms of the category of 'mind'. At the end of the nineteenth century that was no mere archaism but a line of thought that was strongly supported by contemporary Darwinism. If the human species had been produced by a process of biological evolution, it could hardly be the sole possessor of the whole range of qualities regarded as 'mental'. Lower forms of mentality must be present in sub-human species. In fact, the flourishing of comparative psychology towards the end of the nineteenth century was directly due to the exciting incentive held out by Darwinism, namely, tracing an evolution of mental life that would parallel the evolution of physical life.[3]

Such a project had been undertaken by G.J. Romanes with Darwin's blessing. In his *Mental Evolution in Animals* he presented a 'comparison of mental structures' across species, an undertaking he considered to be analogous to the 'scientific comparison of the bodily structures of organisms' carried out by comparative anatomy (Romanes, 1884: 5). But there was a problem. Bodily structures could be studied by any competent observer, whereas mental phenomena were observable only by the single individual in whose mind they took place. Romanes, in common with virtually all Anglo-Saxon writers on this subject, took it for granted that knowledge of minds other than one's own was based, not on observation, but on inference: 'Therefore all our knowledge of mental activities other than our own really consists of an inferential interpretation of bodily activities' (Romanes, 1884: 16). The study of mental evolution in animals would thus have to rely on inferences from bodily activities that could be regarded as furnishing a so-called 'criterion of mind'.[4]

For Romanes and the other comparative psychologists of the time it was quite clear that the problem of animal minds was only a special case of the problem of other minds in general, whether human or otherwise. It was not merely knowledge of animal minds that had to be based on inference from bodily activities, but knowledge of all minds other than one's own. There might well be special risks and difficulties in making inferences about animal, rather than other human, minds, but this was a difference of degree, not a matter of principle.

Underlying the notion of a bodily 'criterion of mind' was the widespread belief that knowledge of one's own and of other minds rested on different foundations. Only one's own mind could be directly *observed*; other minds had to be *inferred* from observations of bodily activity. Such notions were part and parcel of the empiricist tradition whose emergence was discussed in Chapter 3. They depend on the equation of mind and consciousness and on the Lockean division between two sources of knowledge: sensations and reflection on sensory content. Once mind is conceived as something that can only be known directly through some kind of 'inner sense', or consciousness, then other minds can only be known indirectly through reflection or inference.

Because of the continuing cultural embeddedness of these beliefs it is appropriate to note in passing that this is not the only way in which knowledge of self and other can be conceived. For example, in Continental Europe there was a strong tradition of 'expression psychology' that was based on the idea of inherent links between the outer and the inner aspects of mental life which made directly *shared* experience possible, at least among members of the same species. Instead of beginning with the assumption that experience is basically a private affair, one can begin with the notion that experience is essentially something shared. Then one would regard the partial privatization of experience as the outcome of a process of development that requires explanation. Conversely, if one begins by assuming that experience is necessarily private, it is shared experience that requires explanation. Undoubtedly, implicit assumptions of this kind depend very much on a social and cultural context that is mediated both by intellectual tradition and by the conventions of everyday relationships. In a highly individualistic culture, or sub-culture, experience is usually defined as a basically private event.

Certainly, this was the definition accepted without question by the British comparative psychologists and their American counterparts. The Darwinian project of tracing the evolution of mind was therefore a peculiarly difficult one. Because mind was equated with private individual consciousness, it was inaccessible to others, except by inference from its 'criterion'. Moreover, this inference would have to operate on the basis of observers' experience of their own minds, because that was the only experience of mind anyone could have. Morgan, who followed Romanes, was perfectly clear about this: 'Introspective study must inevitably be the basis and foundation of all comparative psychology,' he advised (Morgan, 1894: 37). The reason is that 'man is forced to interpret the psychology of animals in terms of human psychology, since with this alone he has first hand acquaintance through the study of the nature and sequence of his own mental processes' (Morgan, 1894: 38–39).

The attempt to make the private Lockean mind fit into an evolutionary framework was doomed to failure. But this failure had far-reaching consequences. Initially, it promoted an emphasis on the distinction between conscious mind, which was private, and a new category comprising what Romanes called its 'criterion' and Morgan at first called its 'objective manifestations'. This latter category was made up of 'bodily activities'; not all bodily activities though, only those from which inferences about mind might be made. The procedure of comparative psychology now involved three steps. First, one observed one's own mind, then one observed the bodily activities of the other, and finally one interpreted the latter in the light of the former. It must be emphasized that Morgan, and apparently most of his audience, believed that this was how all knowledge of other minds was arrived at, not just knowledge of animal minds.[5]

But in the case of animals, especially lower animals, the tenuous nature of the inferences made tended to become rather obvious.[6] The class of bodily activities that had served as the 'criterion' of mind, or as its 'objective manifestations', increasingly attracted attention in its own right. Morgan began referring to this class as the 'behaviour' of the animal. Not all bodily activity was behaviour; the beating of the heart or the processes involved in the regulation of body temperature, for instance, were not. The kinds of activity that qualified tended to be those that had served as the criteria or manifestations of mind: approach and withdrawal, discrimination, learning, and so on. One could study these and leave out the troublesome task of mental inference. Increasingly, comparative psychologists found it convenient to avail themselves of the category of 'behaviour' when dealing with these activities. This term was suitably non-committal about the mental aspect, expressing the ambivalence and confusion of most practitioners in regard to the deeper issues involved. By common consent, the study of behaviour was 'objective', whereas the introspective study of mind was not. However, most comparative psychologists at this time explicitly declared that the category of behaviour did not exclude mind. Even Jennings, more objectivist than most, believed that the 'processes of behavior' included 'thought and reason' (Jennings, 1906: vi).

A different conception of mind was beginning to emerge here. The equation of mind with a private individual consciousness was proving to be ever more problematical in the field of animal psychology.[7] But when one considers most of the examples of 'behaviour' that were studied among the higher animals one realizes that this category did not just refer to physical movements. Terms like 'learning', 'discrimination', 'reward', 'habit', etc. were not terms that anyone used to refer to physical displacements and nothing more. They were terms derived from the language of human action, a language on which the philosophical distinction between private mind and public movement had never made much impression. The category of behaviour allowed the comparative psychologists to use this language while paying their respects to an ideal of scientific objectivity. They could also claim to be studying psychological processes like reasoning, memory, perception, and so on, insofar as these processes manifested themselves in animals' actions. The old polarity between physical movement and mind as consciousness was no longer tenable. Once in possession of the category of behaviour, comparative psychologists need no longer be concerned with *inferred* minds, though not all of them immediately accepted that conclusion. The point was that the classes of animate motion which provided most of the content for the category of behaviour already incorporated qualities such as goal-directedness, organization and adaptability that had been attributed to the influence of a separate private mind under the old scheme. With their adoption of 'behaviour' comparative psychologists were on the way to shedding a conceptual framework that presented them with insoluble problems.

One way of resolving these problems without shedding the traditional framework was to adopt a thoroughgoing materialism and declare mind an 'epiphenomenon', a kind of emanation given off by material processes that lacks all causal influence. This is what had been entailed by T.H. Huxley's notion of the 'conscious automaton'. Similar ideas had been developed by D.A. Spalding who performed experiments on animals and established radical materialism as a possible, though unpopular, position to take in comparative psychology (Gray, 1967, 1968).

Of far greater importance for the subsequent history of 'behaviour' was the version of materialism represented by the biologist Jacques Loeb, who greatly impressed the young J.B. Watson at the University of Chicago. Loeb's materialism was not simply a matter of philosophical preference but was intimately linked to his deep commitment to what has been called 'the engineering ideal in biology' (Pauly, 1987). That ideal had its roots in the agricultural biology taught in nineteenth-century Germany, but Loeb transformed it into a guiding principle for all biological research. The goal of such research should be the control of existing forms of life and the construction of new forms. Applied to the behaviour of organisms, this principle led him to the study of tropistic reactions and a rejection of theorizing about hypothetical internal processes, whether mental or physiological. To control and alter the reactions of organisms it was important to study their dependence on external, not internal, conditions. The most direct kind of control could be exercised through the effects of external physical conditions on the physical reactions of organisms. From the engineering perspective organisms were regarded as infinitely malleable. They lacked any capacity for 'voluntary' action, so any consciousness they had would be no more than an epiphenomenon, devoid of causal significance. In Psychology, this point of view would become crucial for the concept of behaviour popularized by the movement known as behaviourism. But before we take this up, we need to take note of the broad significance of animal behaviour studies on the eve of that development.

A naturalistic social science

The adoption of the category of behaviour by comparative psychologists would have remained a minor topic in the history of science had it not been for the fact that this field occupied such a significant place in a certain conception of social science. Around the turn of the century there was a growing interest in applying a scientific approach to social problems, and for many this meant erecting a social science on a biological basis. From this point of view, comparative psychology occupied a key place among the sciences. It provided the crucial link between biological and social science, and its investigations were expected to yield knowledge of basic principles that also operated in human conduct.

This generalization of knowledge about animals to knowledge about humans in society depended on the use of the same categories to describe and explain the activity of animals and the activity of humans. Among the most important of these categories were those of instinct, habit and behaviour. If one spoke of a rat following a familiar path and the conduct of humans at work as both being instances of 'habit', one had already established a strong presumption that these were essentially the same kind of thing and therefore to be explained in the same kind of way. Similarly, the device of extending the reference of 'behaviour' from animal activity to human conduct was attractive to those who had developed a faith in the promise of a biologically grounded social science. Subsuming animal activity and human conduct under the same category of 'behaviour' helped to establish the plausibility of the claim that both could be explained by the same 'laws' or principles.

This is quite apparent in the early, pre-behaviourist, texts with titles referring to 'the study of behaviour', 'the science of behavior', or 'the laws of behavior'. McDougall's (1912) goal was the explanation of human social conduct by means of a finite list of instincts that humans shared with animals. He drew freely on recent studies of 'animal behaviour' to establish his claim that human psychology, defined as 'the study of behaviour', was subject to the same instinctive determinants that operated in animal behaviour. Parmelee (1911) presents 'the science of behavior' in terms of a conventional positivist hierarchy that goes from biological foundations via psychology to sociology. Describing what happens at all three levels in terms of the category of behaviour helps him to establish the claim that the same 'elements' are involved at all three levels. Moreover, he promises that the use of the category of behaviour will perform the same service for human psychology and sociology that it has already performed for animal psychology – it will make them scientific and objective. This consideration is also important for Max Meyer's treatise, *The Fundamental Laws of Human Behavior* (1911), though he looks to physiology rather than animal behaviour for inspiration. Although they differ in their specific agendas, all three authors employ the category of behaviour to establish the plausibility of one or other variant of biological reductionism and to reinforce their claim to have adopted a truly scientific approach.

The biological connection was also important for many psychologists who were closer to the mainstream of American Psychology than these three authors. Representatives of the functionalist school were particularly committed to building a science of Psychology on a biological basis and were therefore greatly interested in the categories of comparative psychology. In an earlier programmatic statement, their spokesman (Angell, 1907: 69) had already expressed his enthusiasm for 'the rejuvenation of interest in the quasi-biological field which we designate animal psychology', and which he proclaimed to be 'surely among the most pregnant which we meet in our generation'. A few years later he focused more specifically on the implications of a biological orientation for the categories of a functionalist psychology, the category of behaviour now playing the central role (Angell, 1913). Noting with approval the 'intelligent apprehension of animal behavior' by comparative psychologists and 'the revolt against the domineering claims of introspection' within psychology, he considered that for functionalists like himself it was 'easy to welcome a category like behavior'. Speaking for this psychological school, he continued: 'Allowing for some conservative reservations, its general tendency would be in the direction of sympathetic acceptance of the behavior concept as a general term under which to subsume minor distinctions in modes of action whether conscious or unconscious' (Angell, 1913: 258–259). Such was the functionalist position on 'behaviour' on the eve of Watson's behaviourist manifesto.

By 1913 'behaviour' was firmly entrenched in an expanding field of discourse that employed the same set of categories for elucidating biological, psychological and sociological issues. Those who participated in this discourse did not see eye to eye on the specific biological categories that held the key to the explanation of human social action. Some gave that distinction to the category of instinct, others to adaptation, still others to reflex. But it was the category of behaviour that was to provide the most convenient general framework for all such discussion. Its promiscuous use expressed an assumption that common principles, spanning the biological, psychological and social levels, were indeed waiting to be discovered. It only remained to identify the right ones.

Moreover, because of its origins among a group of biologists particularly concerned with problems of objectivity, 'behaviour' had acquired scientistic connotations. In stating that behaviour was the object of one's investigations one was also proclaiming one's allegiance to 'science' in the Anglo-American sense, that is to say, natural science and not humanistic science or *Geisteswissenschaft*. Far from being a neutral category, 'behaviour' had become a preferred vehicle for those who shared the conviction that human problems would be solved only by adopting a natural science approach.

These connotations of 'behaviour' converged rather smoothly with certain tendencies that were characteristic of the social sciences in the US. From its emergence in the nineteenth century American social science had been marked by an estrangement from history (Ross, 1991). There was a strong preference for naturalistic rather than historical explanation and a tendency to analyse the present in terms of timeless natural laws rather than particular historical constellations. Social Darwinism represented an earlier manifestation of this tendency, but the pervasive 'evolutionary naturalism' (Cravens, 1978) which took its place was a more effective variant in the long run. Functionalist psychology, which sought to explain all aspects of mental activity in terms of their adaptive value, was very much part of a trend that consistently looked to nature rather than to history for its explanation of human conduct. The category of behaviour provided the perfect vehicle for this fundamental commitment, for it provided a way of describing human conduct that insinuated an essential similarity of human and animal activity. It provided the means for discussing human conduct in a way that abstracted from its moral, political and historical dimensions. In the writings of the friends of behaviour these dimensions frequently had ceased to exist and at best played a non-essential role as added on elements. What was fundamental and essential for understanding human conduct was its link to nature, exemplified in animal behaviour.

Isms and ambiguities

In 1913 the 'ism' was added on to behaviour. What effects did this have on a category that was by then quite well established in certain parts of social and biological science? Most obviously, it imparted a distinctly *rhetorical* quality to the term. Once 'behaviour' had been inscribed on the flag of a *movement*, it became a quasi-political token, a focus for the internal conflicts of disciplines as well as for popular attention and argument (Birnbaum, 1965). In the social sciences one could use 'behaviour' as a battle cry to proclaim one's own militant allegiance to the cause of science and the value of objectivity while denigrating one's opponents as old fogeys held in thrall by yesterday's superstitions. In the popular media one could package all sorts of nostrums as 'behavioural' in order to help one's sales pitch.

Within Psychology, 'behaviour' now became an *arena* in which theoretical conflicts were fought out. These conflicts often took the form of wrangles about what was and what was not to be included in the category of behaviour, about what exactly constituted a 'behaviourist', and about the nature of the 'laws' that were supposed to 'govern' behaviour. The last of these issues quickly became entangled with the fate of the category of 'learning' which is addressed in a later section of this chapter. The question of what it took to call oneself a behaviourist is relevant in the present context only insofar as it illustrates the way in which 'behaviour' had become an arena in which various theoretical issues were fought out. One of these was the issue of introspection. Years before the advent of behaviourism, American psychologists had shown a clear preference for non-introspective data (Cattell, 1904; O'Donnell, 1985), but many of them resisted the idea that such data was to be ruled out of court as a matter of principle. Perhaps one could allow it back in as 'verbal report' and still define psychology as 'the science of behaviour'. It soon became clear that 'behaviourism does not stand or fall with any special hypothesis which behaviourists happen to like' (Woodworth, 1924: 263). Increasingly, it functioned more like a framework that incorporated many of the unquestioned assumptions and unquestionable values of American psychologists. Within this framework there was room for arguing about details.

From the beginning of behaviour's career as a quasi-scientific category there was disagreement about its meaning. Early on, that meaning was closely tied to its relationship to the category of 'mind', as we have seen. Behaviour was that from which mind could be *inferred*. But when the mind was that of an animal, this became problematical. Yet, most of the pioneers of animal psychology remained convinced that mind was somehow *manifest* in behaviour. The question of how this was possible was one that intrigued several American philosophers. E.A. Singer (1912: 209) proposed that 'our belief in consciousness is an expectation of probable behavior based on an observation of actual behavior, a belief to be confirmed or refuted by more observation'. That amounted to a reinterpretation of the problem from a novel, *pragmatist*, perspective. The issue is no longer one of the real existence of mind as an object, but of the foundations for one's belief in such a thing. From this vantage point two lines of development were possible. One could go on to explore the social basis for the attribution of mental qualities. Alternatively, one could revert to a realist position and redefine mind in terms of particular objective qualities in behaviour. The New Realist philosophers, more influential than Singer, took the second road.

For E.B. Holt, (1915: 166) behaviour was an activity of the whole organism that could be 'shown to be a constant function of some aspect of the objective world'. What distinguishes behaviour from mere physical movement is not its accompaniment or activation by a mental event but its functional relationship to some 'objective reference' outside itself. Behaviour is movement that varies with changes in the objective circumstances. That is the difference between a stone rolling from point *a* to point *b* and an organism moving *towards* the light, for example. Intentionality is not due to a mind *behind* the movement, it is discoverable *in* those movements that are behaviour. That, briefly, was the foundation on which Holt's student, E.C. Tolman (1932), was to construct his notion of 'molar' behaviour, to be contrasted with 'molecular' behaviour that consisted of nothing but a chain of muscle contractions. The latter was said to have been the notion of behaviour with which Watsonian behaviourism operated.

But in fact, both Watsonian and molar behaviourism recognized either conception of behaviour, depending on the rhetorical context. When they wanted to emphasize their credentials as natural scientists the molar behaviourists would insist on the physical basis of molar behaviour, and when Watsonian behaviourists referred to the practical applications of their theories, that were so important to them, they had to speak of actions, like memorizing, drawing, writing, rather than physical movements. As many have observed (see Kitchener, 1977; Lee, 1983), behaviourism's use of the category of behaviour has always been marked by *ambiguity*. The reason for that lies in the basic incompatibility of the twin goals of the movement. Behaviourists desperately wanted to be recognized as natural scientists, and, to the degree that their image of natural science included good old-fashioned mechanistic materialism, they would emphasize that behaviour was ultimately reducible to physical movement. But they also wanted to build a practical science of social control, and that had to operate with purposeful action rather than with physical movement. It was only when the crude physicalism of an earlier notion of science lost its cash value in enlightened circles that the 'molecular' conception of behaviour faded into history. But by then the goal of 'prediction and control' had come to dominate the discipline in any case.

The practical exemplar

During the heyday of behaviourism the transformation of 'behaviour' into an arena also took the very concrete form of the rat learning experiment (pigeons came later). It was in the construction and interpretation of such experiments that fundamental theoretical issues were supposed to be decided. In effect, these experiments functioned as prototypical situations that imposed a very specific meaning on an otherwise almost meaningless term. To a prominent experimentalist the essence of the behaviourist programme seemed to lie in the fact that 'it proposed to take the animal experiment as the model for experiments on human beings' (Woodworth, 1924: 260). That was during the early period of 'classical' behaviourism. But during the subsequent period of neo-behaviourism this aspect of the programme was realized on a scale that Woodworth probably had not foreseen. What consequences did that have for the meaning of 'behaviour'?

Obviously, it would reinforce the association of 'behaviour' with the already existing and pervasive tendency to explain human social conduct in naturalistic terms. But more specifically, the link of 'behaviour' to the prototypical animal learning situation reinforced the understanding of behaviour as something that organisms did as *individuals*. Indeed, the typical animal learning experiment represents a particularly stark depiction of conduct as the affair of independent individuals. What is studied in these experiments is the intercourse of single and isolated animals with their physical environment. These animals represent individuals even more drastically thrown on their own resources than the lone gunmen of the mythical Wild West.

Moreover, such experiments provide an ingenious way of eliminating the element of human communication that is such a troublesome yet absolutely necessary part of experiments with human subjects. In their interpretations of experimental data psychologists had traditionally tried to ignore the fact that all such data were the product of a social interaction (Danziger, 1990a). In animal learning experiments, however, they had managed to construct a situation in which human communication had been eliminated once and for all. The subjects whose behaviour provided the prototype for behaviour in general were speechless subjects. They lacked the means for participating in a language community with their controllers or anyone else. 'Behaviour' was basically an attribute of singular, non-communicating individuals. Applied to humans, anything social or cultural could enter this world only in the form of 'stimuli' external to the individual. No conception of community beyond that of an aggregate of individuals could be formulated on this basis.[8]

'Behaviour' is an extremely abstract category that depends for its meaning on the use of exemplars that function as meaning-conferring prototypes. This is what the animal learning experiments of the neo-behaviourists supplied. It was in these experiments that they sought the key to the principles that governed behaviour in general; here, as nowhere else, behaviour was supposedly manifested in its pure form. So far from that being the case, however, these experimental situations produced exemplars that were pure cultural artefacts.

A cross-cultural comparison will help to illustrate this. European psychologists generally treated behaviourism as a bizarre American fad that was not to be taken seriously. Only their forced emigration obliged some of them to alter that stand. But there was one prominent exception in the 1920s. Karl Bühler, head of the Vienna Psychological Institute and President of the German Psychological Society between 1929 and 1931, agreed that the behaviourist programme was *echt amerikanisch* (Bühler, 1927: 20), but accepted that the study of animal behaviour, or conduct (*Benehmen*), as he called it, provided knowledge that must form part of the foundations of psychology. Accordingly, he tried to incorporate the behavioural aspect in his own synthesis. But what did he take as the prototype of behaviour for this purpose? Not the solitary rat in the maze, not even the problem-solving primate, but the communicating bee! It was the work of von Frisch on the 'language of the bees' that provided Bühler with the exemplars he was looking for. That was because he took it for granted that the kind of animal behaviour that would be most relevant to human behaviour would have to be *communicating* behaviour. For Bühler, humans were fundamentally communicating creatures, and the psychological work of his mature years was mostly devoted to the study of language (Bühler, 1990).[9] The point here is not whether Bühler was right and the rat runners were wrong, but that the choice of the exemplars which give meaning to abstract 'behaviour' depends on culturally determined preconceptions about the fundamental nature of human conduct.

Behaviour and control

Perhaps the deepest and most permanent effect of the behaviourist annexation of the category of behaviour is to be found in the formation of a close association between behaviour and 'control'. That association is already present in the Watsonian concept of behaviour which had embodied the 'engineering ideal' personified by Jacques Loeb in Biology. For Watson, the goal of psychology was not the understanding of the mind, but 'the prediction and control of behaviour': 'If psychology would follow the plan I suggest, the educator, the physician, the jurist, and the business man could utilize our data in a practical way, as soon as we are able, experimentally, to obtain them' (Watson, 1913: 168). The areas of Psychology which were flourishing, according to Watson, were those that had effectively adopted his goal, for example the psychology of advertising, of drugs, and of tests, as well as 'experimental pedagogy', legal psychology, and psychopathology. These examples illustrate that Watson's category of 'behaviour' was quite broad. Whatever could be reliably predicted and placed under external control could be included. The human engineering perspective, which Watson made explicit, had in fact been prominent in so-called applied psychology for some years. One effect of his campaign was to establish a firm link between that perspective and the category of behaviour.

By the 1920s that link had become unmistakable. This was a time when American Psychology became a significant beneficiary of large scale private funding available for research that might contribute to the control of social problems (Samelson, 1985). There was a clear shift away from the old style of psychological research on the individual mind to research on the distribution of dispositions and on malleability in large groups of subjects (Danziger, 1990a). It now became quite generally accepted that research ought to produce the type of knowledge that would be immediately or potentially useful for purposes of social control. To distinguish this kind of knowledge from the painstaking investigation of consciousness that had characterized an earlier phase of modern psychology, the apostles of the new wave mostly followed Watson's lead and proclaimed their interest in 'behaviour'. The term began to function as a kind of banner under which all those who believed in a science of social control could rally. By 1925, one observer saw a 'Kingdom of Behavior' that had resulted from 'a shift from understanding to control', to 'management, direction, betterment, greater effectiveness' (cited in Samelson, 1985: 42). Those who declared the subject matter of Psychology to be 'behaviour' almost always linked this to the stipulation that its goal was to be the 'prediction and control' of this behaviour.

Behaviourism had been revolutionary in the sense that, although it had sprung out of a discourse about the place of mind in nature, it had transformed that discourse into one about the control of human conduct. As behaviourism matured and its discourse came to dominate the discipline, the polarity of mind and behaviour became more and more peripheral. Eventually, terms like 'cognitive behaviour' and 'cognitive behaviourism' were accepted with hardly a raising of eyebrows. The point was that the discursive context which 'behaviour' signalled had by then changed from questions about mind to questions about techniques for the control of human conduct. 'Behaviour' had come to mean any aspect of human activity that could be predicted and controlled by psychologists.

The neo-behaviourism of the 1930s and 1940s completed this transformation. It marked the formalization of the *positivism* that had always been implicit in behaviourism. Watson had ejected references to brain physiology almost in the same breath with the expulsion of the mind. Behaviourism meant not penetrating beyond the surface of the organism, sticking to what could be directly observed and controlled. Neo-behaviourism formalized the primacy of the observable and manipulable by adopting the language of independent, dependent, and intervening variables. That language is discussed at some length in Chapter 9. It was based on an equation of scientific knowledge with the technical ability to produce specific effects. Psychological constructs now had to be defined in terms of specific operations producing specific effects.[10] But only operations carried out by qualified psychologists in accordance with the prevailing methodological norms of the discipline were recognized for this purpose. Adopting a 'behavioural' approach came to mean that one was committed to these norms and to the controlling role of the scientists who acted in accordance with them.

The older psychology had been obliged to defer to the privileged access to their own conscious experience by subjects in psychological experiments. In 'behavioural science', it was the scientist who was privileged. His/her operations defined the phenomenon under investigation, and they defined it in the service of the goal of 'prediction and control' of the subject's reactions. In this context, 'behaviour' had come to mean an activity defined in such a manner as to make it an object of behavioural science. It referred to that which was predictable and controllable by the specific means in vogue among a particular group of experts.

The abstraction of 'learning'

One of the curious features of the original behaviourist programme was the contrast between the vagueness of its positive proposals and the very specific targets of its critique. Its objections to introspection were backed up by specific examples from the psychology of thinking and imagery, but there was little positive theoretical content, beyond some forcefully expressed philosophical preferences for mechanistic and materialistic explanations.

Soon the concept of 'conditioning' was enthusiastically adopted as an appropriate filler for what looked suspiciously like a theoretical vacuum. This was a concept that had been derived from work that the Russian physiologist, I.P. Pavlov, had done on conditioned reflexes. There was however a world of difference between the meaning that this work had had for Pavlov himself and the significance it acquired in the discourse of American behaviourism. For Pavlov, work on conditioned reflexes was a *means* of generating data from which hypotheses about cerebral processes in intact animals could be derived. For the behaviourists, conditioning became an *explanatory category* that would suffice to account for adaptive behaviour without any reference to brain physiology. Pavlov was well aware of this difference. In an article published in America, entitled 'Reply of a physiologist to psychologists', he expressed himself as follows:

> The psychologist takes conditioning as the principle of learning, and accepting this principle as not subject to further analysis, not requiring ultimate investigation, he endeavours to apply it to everything and to explain all the individual features of learning as one and the same process. . . . The physiologist proceeds in quite the opposite way. . . . Conditioning, association by contiguity in time, conditioned reflexes, even if they serve as the factual point

of departure of our investigations, are nonetheless subject to further analysis. We have before us an important question: What elementary properties of brain-mass form the basis of this fact? (Pavlov, 1932: 91–92)

Conditioning of the Pavlovian, or 'classical', kind became the first of a series of pseudo-explanatory principles introduced to provide the theoretical content that the behaviourist framework in itself could not supply. But what was it that conditioning and its successors were meant to explain? The answer was 'learning'. Prior to the 'theories of learning', which made up the specific theoretical content of neo-behaviourism, there had to be a class of phenomena grouped together in the category of 'learning'. Moreover, there had to be a conviction that this was not a psychologically trivial category, but one of central importance for achieving the goal which the discipline had set itself. Indeed, it was widely believed that the 'laws' of 'learning' represented the fundamental principles of a scientific psychology. But this was a belief that relied on a very recent historical construction, namely, a category of learning phenomena that was sufficiently unified to be explicable in terms of a single set of regularities. In Pavlov's words, basing everything on learning meant 'to represent all the separate sides of learning as one and the same process'.

In the early days of modern psychology such an idea would have seemed preposterous. There was as yet no category of learning-in-general subject to one set of laws. There were a number of contexts in which the term 'learning' cropped up, but, at best, they were linked by only the vaguest analogies. Strangely enough, vague analogies were sufficient for the construction of an abstraction that was to dominate theoretical debate in mid-twentieth-century American Psychology. How did this strange construction originate?

Of course, casual references to children 'learning' such things as walking and talking, or adults 'learning' to play a musical instrument or a foreign language, go back a long way. But such references carried no implication that all examples of learning constituted one natural kind united by common features. On the contrary, in pre-twentieth-century texts of mental philosophy or psychology such casual references to learning were always embedded in discussions of those categories of phenomena that were considered significant, categories like association, habit, imitation, memory, education, training, and so on. It was these latter that provided the discourse with its structure and focus, not any category of 'learning'.[11] That relationship was reversed in twentieth-century psychology, so that these other categories now became mere examples or applications of the more fundamental category of 'learning'. This reversal could not have occurred if 'learning' had not acquired a significance it did not previously possess. How and why did this happen?

At the beginning of the century there were three entirely different contexts in which 'learning' was beginning to be accorded a new significance. One was comparative psychology, the second was the acquisition of skills, and the third was educational psychology. We will consider these in turn.

In evolutionary comparative psychology, the *activity* of learning had been singled out as a crucial 'criterion of mind' (see 'Mind as inference', p. 89ff.). Romanes (1884: 20–21) wrote as follows:

> The criterion of mind, therefore, which I propose . . . is as follows: – Does the organism learn to make new adjustments, or to modify old ones, in accordance with the results of its own individual experience?[12]

His argument was based on the belief that learning from experience must involve 'choice', and choice necessarily implied the presence of 'conscious intelligence'. Be that as it may, studies of the learning of new adjustments became part of evolutionary comparative psychology. Learning had acquired some significance. However, its role was still a secondary one; it was merely a criterion of things like mind and intelligence which were truly fundamental. In this context, 'learning' did not refer to a mechanical, but to a conscious, process.

Subsequently, biologists, such as Loeb and Jennings, who worked with lower organisms, became interested in a basic *modifiability* that seemed to be present in the behaviour of even the simplest animals. But at this level consciousness was effectively out of the picture. One was dealing with automatic adjustments whose physico-chemical basis could sometimes be demonstrated. Sherrington's work on postural adjustments based on unconscious reflex mechanisms pointed in a similar direction. There was a great deal of variety, and even disagreement, regarding the exact nature of the automatic adjustive responses of organisms, but it was clear that a category of such phenomena had to be recognized.

The question was whether this kind of modifiability had much significance for the higher organisms, especially humans, and whether it had much significance for more 'intelligent' levels of response. Biologists differed on this, and the position they took had much to do with their attitude to the theory of evolution. Clearly, the hypothesis that the modifiability of animal behaviour involves the same mechanisms at all levels of evolutionary development restricts the scope of evolutionary theory. On the other hand, the hypothesis that processes of adjustment changed significantly in the course of evolution enhances the domain of evolutionary theory. This was the position of Darwinians like Romanes. Loeb, on the other hand, did not have much time for evolutionism (Pauly, 1987). His goal for biology was as a science providing the principles for the control of life, and in his day evolutionism seemed to have no contribution to make to that. By comparison, the study of primitive adjustive mechanisms seemed much more promising, and Loeb proceeded to push the applicability of these studies to higher forms of life. His claim – really no more than a hope – was that the same fundamental principles of modifiability applied at all levels of animal development.

The second context in which questions about the nature of 'learning' were being raised involved studies in the acquisition of practical skills. American psychologists took an early lead in this area, publishing data on the acquisition of such skills as telegraphy (Bryan and Harter, 1899), shorthand (Swift, 1903), and typewriting (Swift, 1904). Quite remarkably, although these were all instances of learning how to communicate through new media, the process was not conceptualized as social but only as involving changes within a singular individual. These changes were, however, treated as instances of a significant class of intra-individual events. For example, in tracing the acquisition of skills in telegraphy one was supposedly studying something much more significant, namely, 'the acquisition of a hierarchy of habits' (Bryan and Harter, 1899). The concept of 'habit' had an impressive pedigree, having played an important role in empiricist mental philosophy since the days of Hume. In the nineteenth century the motor component of habits had received much attention in the wake of the expansion of the reflex concept.[13] For Carpenter (1874), and for William James who followed him closely, habits were essentially sensori-motor adjustments that also acted as the repository of social custom. Studying 'the telegraphic language' as an instance of habit acquisition (Bryan and Harter, 1899) fitted perfectly into this tradition.

Nevertheless, the transformation of the topic of habits into a label for laboratory studies of practical skills entailed a certain shift of emphasis. The older empiricist discourse on habits had been more concerned with their general structure than with the specific details of their acquisition. Laboratory investigations inspired by a practical interest in what Bryan and Harter called 'the psychology of occupations', however, would necessarily be looking at the details of habit acquisition. They would 'portray the actual typical procedures of men in learning or in failing to learn' (Bryan and Harter, 1899: 349). Quite subtly, the main focus had begun to change from the nature of 'habits' to the nature of 'learning', the process involved in their formation. This quickly becomes more explicit in subsequent investigations of practical skills. They are now identified as contributions to 'the psychology of learning' (Swift, 1903, 1904). In due course, there is talk of something called 'the learning process' (Swift and Schuyler, 1907; Richardson, 1912), whose common features show up in a variety of skills.

Thirdly, 'learning' is something that young scholars and students do in educational settings. But how does one define the psychological issue here? Traditionally, the pupil's learning had been seen as incidental to the real issue, which was that of memory. That was one of the few truly ancient psychological topics, and speculation about the nature of memory had existed for centuries. But modern practices brought a change in the relative status of memory and learning. This was because the most widely adopted programme for the experimental investigation of memory (Ebbinghaus, 1885) defined memory in terms of the *work* of memorizing and not in terms of the *experience* of remembering.[14] In this context 'learning' was used as a synonym for memorizing, and experimental investigations were designed to answer questions about the relative efficiency of different techniques of learning.

That approach furnished an excellent basis for a major programme of applying psychological knowledge in school settings. By the beginning of the twentieth century there was an interest in the rationalization of educational practice in all the rapidly industrializing countries of the northern hemisphere. Traditional methods and goals of education were increasingly being rejected as inappropriate for modern conditions. Schooling ought to prepare individuals to occupy workplaces in an industrial society governed by the goal of maximizing efficiency. It would not be able to do this very effectively, unless the same principles of technical rationalization that were being so impressively employed in industry were also put to work in reforming educational practice. Such was the background for the work of a group of German educators who were committed to putting pedagogics on a scientific footing. That meant conducting experiments to discover the relative effectiveness of different techniques of learning, a procedure for which Ebbinghaus' studies of memory had laid the basis. The central figure in this endeavour was the psychologist Ernst Meumann, whose standard text for the area had gone into a second edition by 1908 under the title 'Economy and Technique of Memory' (Meumann, 1908). The third edition appeared in English as *The Psychology of Learning* (Meumann, 1913). By then, there was an established area of psychological research devoted to the comparison of different techniques of learning from the point of view of efficiency.

The German work in this area evoked considerable interest in America, where similar lines of experimentation were beginning to open up. In many ways, E.L. Thorndike was Meumann's counterpart in America, using his academic position at Teachers' College, Columbia, to promote studies of the effect of various conditions on the learning of school-like tasks. In 1913 his three-volume *Educational Psychology* was published, the second volume of which was entirely devoted to *The Psychology of Learning* (Thorndike, 1913). The third context of learning studies was by now well developed. Thorndike's volume included a nine-page bibliography, though some of the items referred to research on animal learning and on the acquisition of skills.

In this respect there was a striking difference between Thorndike's text and Meumann's text. The latter was exclusively devoted to human memorizing of symbolic material. In Thorndike's case, however, there is an introductory chapter on 'the laws of learning in animals', as well as frequent reference to research on the acquisition of skills. Closer inspection reveals that the meaning of 'learning' is quite different for these two authors. For Meumann, 'learning' was essentially a synonym for 'memorizing'. It was an intentional human activity, involving conscious attention, and studied in the very specific context of school work. Thorndike's 'learning' was a far grander affair. What went on in learning at school was only one manifestation of a biological process that could also be observed in animals – monkeys, cats, chicks and turtles being explicitly mentioned. Moreover, this process obeyed a few simple 'laws', which Thorndike enumerated. The same laws operated in the acquisition of human motor skills and in ideational learning. What had been studied in three very different contexts was now assumed to be all the same process. Clearly, a psychological phenomenon as pervasive as Thorndike's 'learning' would provide a focus for a significant part of the discipline. By contrast, Meumann's 'learning' was merely a technical means employed in a very restricted practical context. Neither he nor any other European psychologist had thought of 'learning' as defining a major area of basic psychological research and theory.

But in America Thorndike was only the most influential contributor to an emerging discourse that took 'learning' in the grand sense as its object.[15] His had not been the first volume to be devoted to the topic. A text on *The Learning Process* (Colvin, 1911) had appeared two years earlier, claiming that the process was similar in animals and children and also including some reference to the acquisition of skills. In the year that Thorndike's text was published, the *Psychological Review* carried an article on 'The laws of learning', based on the proposition that 'the several processes which in various animals are called learning, habit-formation, memory, association, etc. are at bottom one and the same process' (Haggerty, 1913). Soon, it seemed perfectly reasonable to undertake experimental studies that used both rats and humans as subjects in research on the learning process (e.g. Pechstein, 1917; Webb, 1917).[16] Psychologists also adopted the practice of pooling data from a number of subjects to produce 'learning curves' based on averages that need not correspond to the performance of a single living organism. Such statistical abstractions often functioned as the practical counterpart of the conceptual abstraction involved in the category of 'learning'.

During the period of Psychology's involvement with the military in World War I the term 'learning' was used as a vehicle for advancing extravagant claims to professional expertise. It was suggested that psychologists had scientific knowledge of 'learning' which would be applicable in a broad variety of contexts, from 'the learning of complicated military manoeuvres' to 'losing one's temper' (Strong, 1918).

The post-war period saw the firm establishment of 'learning' in the abstract as a major subdivision within American Psychology. Introductory texts, beginning with Woodworth's very successful *Psychology* of 1921, would discuss evidence from animal experiments in conjunction with work on human skills and studies of memory to promote the notion of so-called 'laws of learning' whose domain was universal. Earnest debates about the precise content of these laws merely strengthened belief in their existence. Over the next two decades, 'theories of learning' emerged as the focal area of basic theorizing in American Psychology. Always abreast of current trends, Woodworth (1934: 223) writes in the third edition of his text that 'the work of psychology must consist very largely in the investigation of learning'. Moreover, animal experiments, especially on rats, have moved up to become the favourite vehicle for this kind of work. As the text explains: 'Psychology seeks to unravel the learning process not as a specifically human process – which it is not – but as a process common to many if not all animals. The simpler processes in the rat are instructive just because of their comparative simplicity' (Woodworth, 1934: 225).

The fate of the category of 'learning' was closely intertwined with that of behaviourism. Watson (1914: 45) seems to have anticipated this: 'On account of its bearing upon human training, learning in animals is probably the most important topic in the whole study of behavior.' The laws and the theories of learning provided the content which behaviourism lacked, an assimilation that was based on common presuppositions. First, there was the implicit minimizing of evolutionary change.[17] By contrast with the Darwinians, the 'laws' the behaviourists were interested in were not historical accounts that explained massive natural changes over long periods of time. Their model was much closer to physical engineering than to evolutionary biology. What they were looking for were principles of wide applicability that could be employed in the practical control of conduct. For that they required abstract categories of manipulable short term change (Mills, 1997), and that is what the theories of 'learning' provided. In this perspective, all organisms were seen as potential objects of control, not as the products of a historical process. Organisms were not only 'empty', devoid of fixed species characteristics, they were also lonely, that is, they were studied in isolation from any natural environment. In the case of humans, that converged with the tendency to perform a radical divorce between individual and society. In this usage, 'learning' was always a phenomenon of individual change, never one of co-change among several individuals sharing a social field.

The categories of 'behaviour' and 'learning' undoubtedly represented the purest expression of twentieth-century modernism within the entire domain of psychology. There is considerable irony in the fact that the categories which most stridently proclaimed Psychology's abstract universalism were precisely the ones which were most parochial in a cultural and historical sense.

Notes

1. More recently, the category of 'cognition' has played a similar role. The idea of abstract laws uniting many domains of psychological functioning, irrespective of content, reappeared in the form of the category of 'cognition' just when 'learning' could no longer play this role effectively. But this development falls outside the time period of this book.

2. The qualification 'relevant' is necessary, because, earlier, one can find instances of talk about the 'behaviour' of physical substances. Certainly, this played a role in the choice of the term by biologists and psychologists, because it was seen as establishing the term's value-free, amoral pedigree. But 'behaviour' was never a technical term in the physical sciences; it became one only when it began to be used in a biological and especially a psychological context.

3. In this context the term 'Darwinism' must be understood to refer to doctrines that were not unique to Darwin but owed a great deal to Herbert Spencer. For the historical background, see R.J. Richards (1987).

4. Romanes did not invent the notion of a bodily criterion of mind. It had emerged out of the Pflüger–Lotze controversy mentioned in Chapter 4 and was the subject of considerable discussion in the latter part of the nineteenth century. William James (1890) puts this topic right at the beginning of his *Principles of Psychology*.

5. One of the first texts to define Psychology as 'the science of behaviour', prior to the advent of behaviourism (Pillsbury, 1911), grounded its definition in this belief. For it, behaviour was the most important of 'the evidences of mind'.

6. The point had been made quite forcefully by Wilhelm Wundt in 1892, when he noted 'the unfortunate absence of the critical attitude' in the work of Romanes (Wundt, 1894: 343).

7. For some details of how this manifested itself in practice, see Mackenzie (1977).

8. It is likely that these features of the category of behaviour helped its promotion as an alternative to 'social' during the McCarthyite period. There were political factors involved in the push to replace 'social science' with 'behavioural science' during this period. (Senn, 1966).

9. It took over half a century for this work to be translated into English! Cultural dissonance played a significant role in Bühler's being written out of the (American) history of psychology (see Brock, 1994; Weimer, 1974).

10. For a thorough and up to date discussion of the historical link between neo-behaviourism and operationism, see Mills (1997).

11. William James' famous chapter on 'habit' in his *Principles of Psychology* (1890) provides an example of this.

12. This was actually a self-quotation from his slightly earlier *Animal Intelligence* (1882).

13. See Chapter 4.

14. For further clarification of this fundamental point, see Danziger (1990a).

15. Thorndike had also been the first to use 'learning' as a category for popularizing psychology. In his early work on animal learning he had used the then current expression 'animal intelligence', but in his popular *The Human Nature Club* (Thorndike, 1902), which followed shortly thereafter, he already uses 'learning' as a psychological category.

16. One reason for the early popularity of the maze as a device for the study of learning may have been that it was a task which could be given to both rats and humans, indicating that one was dealing with fundamentally the same phenomenon in both cases. The maze also seemed to require motor learning, making it easy to put maze learning in the same category as the acquisition of a skill. Thus, studies of maze learning became the investigative practice that provided the perfect empirical content for a category of learning that had been derived by abstraction from three very different contexts.

17. Thorndike (1911: 280) had spelled this out at an early stage: 'If my analysis is true, the evolution of behavior is a rather simple matter. Formally the crab, fish, turtle, dog, cat, monkey and baby have very similar intellects and characters. All are systems of connections subject to change by the law of exercize and effect.'

7

MOTIVATION AND PERSONALITY

Nothing attests to the vaulting ambition of American inter-war Psychology quite as well as the invention of two wholly new fields of investigation – motivation and personality. More than any others, these two categories staked the claim of Psychology *as a discipline* to special and privileged knowledge about the *entire* range of human affairs.

One of the first systematic texts in this area begins as follows:

> All behavior is motivated. Getting out of bed when the alarm clock rings, brushing the teeth, shaving, selecting the day's necktie, ordering rolls and coffee or ham and eggs from the menu card, picking up the paper to read the news – these everyday activities are all causally determined. You take them for granted, generally being unconscious of any motive determining what is being done. Nevertheless a definite motivation is invariably present. (Young, 1936: 1)

Scientific psychology promised to unravel the secrets of this invisible but ubiquitous network of causal determination. No human action, no matter how trivial, could now escape its reach. And what was not covered by the general laws of motivation, namely the differences among individuals in carrying out these actions, would be netted by the new field of scientific personality research. Seamlessly, the new Psychology will account, not just for the 'how' but for the 'why' of all human activity.

In its earlier phase the new experimental psychology had had far more modest aims. It would not have occurred to a Wundt or a Titchener that their field ought to provide explanations of everyday conduct. The special problems arising out of the structure and flow of human consciousness, which they investigated, would impinge on ordinary human activity here and there, but there was no pretence that their laboratory studies would yield knowledge that could, in principle, explain why anyone ever did anything. Accordingly, their modest psychology lacked the categories that would have provided a suitable framework for accommodating such claims. They knew neither 'motivation' nor 'personality'. Baldwin's (1901) comprehensive turn of the century dictionary has no entry for 'motivation' at all. It has an entry for 'person', whose content is entirely philosophical, and an entry for 'personality', whose content is entirely medical, but there is no psychology of personality.